

Attention Is All You Need

Ashish Vaswani et al.

Google Brain 2017

Abstract

We propose Transformer, based solely on attention.

Keywords: attention, transformer

1 Introduction

RNNs suffer from sequential computation which prevents parallelization.

2 Method

The Transformer uses stacked self-attention and fully connected layers.

2.1 Scaled Dot-Product Attention

Input consists of queries and keys of dimension d_k .

2.2 Multi-Head Attention

Linearly project the queries, keys and values to different dimensions.

3 Experiments

On WMT 2014 English-to-German, our model achieves 28.4 BLEU.

3.1 Machine Translation

Our big model outperforms all previously reported models.

3.2 Model Variations

We varied the base model in different ways.

4 Conclusion

We presented Transformer, the first sequence transduction model based entirely on attention.

References

[1] Bahdanau D et al. Neural machine translation. arXiv:1409.0473, 2014.